

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester I Examination, 2022-23

Paper No. CHB-101: Structure & Bonding and Organic Chemistry-I

Time: 3 Hours

Maximum Marks: 70

Instructions: Answer five questions in total, including Question 1 from each section which is compulsory. Use separate answer books for Section A and Section B.

SECTION A

(Structure and Bonding – Marks: 35)

Question 1. Answer all parts of the following:

[5 × 2 = 10]

- (a) What kind of interaction exists among the atoms when helium is liquefied?
- (b) Discuss the Electron Pool Theory in brief.
- (c) What is meant by a node in an orbital?
- (d) Arrange the orbitals 3s, 3d, and 4s in order of increasing energy in the case of the hydrogen atom.
- (e) Why is it necessary to convert the Schrödinger wave equation in terms of polar coordinates for a hydrogen-like atom?

Question 2.

- (a) What is the expected bond order in the Be₂ molecule? How would you explain the fact that beryllium exists as a metal? [6]
- (b) Discuss the solid-state structure of NaCl using a suitable diagram. [6]

Question 3.

- (a) Write the expressions for x , y , and z in terms of polar coordinates for the hydrogen atom. [6]
- (b) Using the VSEPR model, explain the geometries and bond angles of NH₃ and NF₃. [6]

Question 4.

- (a) Derive the Schrödinger wave equation assuming the electron to be a standing wave and explain all the terms involved in it. [6]

- (b) Discuss the bonding in H_2^+ molecule ion on the basis of Molecular Orbital Theory. [6]
- (c) How would you explain the fact that the outermost electron in potassium goes to the 4s orbital rather than a 3d orbital? [6]

SECTION B

(Organic Chemistry-I – Marks: 35)

Question 5. Answer all parts of the following: [5 × 2 = 10]

- (a) Write appropriate reactions for the formation of *cis*-1,2-cyclopentanediol and *trans*-1,2-cyclopentanediol from cyclopentene.
- (b) Draw the energy profile diagram for the following reaction:
- $$(\text{CH}_3)_3\text{C-Cl} + \text{NaOH} \rightarrow (\text{CH}_3)_3\text{C-OH} + \text{NaCl}$$
- (c) Identify whether the chiral center in 2-chlorobutane is *R* or *S* for both enantiomers.
- (d) Give the name and structure of a reagent for the selective oxidation of primary alcohols to aldehydes.
- (e) Explain why the chair conformation of cyclohexane is free of angle and torsional strain.

Question 6.

- (a) Discuss the base-catalysed tautomerisation of ethyl acetoacetate. [6]
- (b) Discuss the reactions of the following with ethyl acetoacetate, showing mechanisms: [6]
- (i) HNO_2
- (ii) NH_3

Question 7.

- (a) Discuss in detail 1,2- vs. 1,4-addition reaction of chlorine on 1,3-butadiene with energy profile diagrams. [6]
- (b) Write in detail about the esterification reaction of alcohols and compare the relative reactivity of primary, secondary, and tertiary alcohols. [6]

Question 8.

- (a) Compare the acidity of ethane, ethylene, and acetylene, and justify the preparation of sodium acetylide. [6]
- (b) Complete the following reactions and write the major product: [6]
- (i) $\text{CH}_3\text{-C} \equiv \text{CH} + \text{H}_2\text{O} \xrightarrow{\text{dil. H}_2\text{SO}_4, \text{HgSO}_4} ?$
- (ii) $\text{CH}_3\text{-CH=CH}_2 + \text{HBr} \xrightarrow{\text{Peroxide}} ?$

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